UNIFORM BOTTOMLESS-RECTANGLE HYPERGRAPHS WITH NO3-SHALLOW HITTING SET

BALÁZS BURSICS, BENCE CSONKA, LUCA SZEPESSY

Editorial note. The counted hyperedges are captured m-point traces of bottomless rectangles, as defined at182-183 and used at470/474; source466 contained means captured in this range-family sense, not every subset of a larger trace.

Editorial note. The word northwest for B_i at464 conflicts with the same paragraph precise inequalities x3<x1<x2 and with the native figure. Those explicit coordinates place B_i to the right of A_i and C_i to the left. The author text and figure are retained.

Editorial note. The groups C_i,j lie above b_i^j and below b_i^(j+1) when j<m; for j=m the precise condition y(b_i^k)<y(C_i,j) iff j>=k puts the final group above all B-points. No nonexistent b_i^(m+1) is needed.

Editorial note. The finite copies above the base points can be horizontally scaled into disjoint small neighborhoods, with every non-anchor point above the base line. This realizes the author separate copies without allowing a witnessing rectangle to capture points of another gadget.

Editorial note. In Case2, the m-3 rightmost A-points exclude the leftmost anchor p_i. The four disjoint consecutive triples in B_i use the first12 B-points when m>12.

Editorial scope. One lower-bound construction for shallow hitting sets in a uniform planar range hypergraph. The native figure is part of the proof. Arithmetic progression colorings, strip hypergraphs, and polychromatic upper bounds are not counted.

Allowed standard prerequisites (editorial).

- Finite planar range hypergraphs. A range captures exactly the input points lying in it. Bottomless rectangles have two vertical sides and a horizontal upper side, with no lower bound.. Verify each explicitly chosen edge has exactly m points.
- Elementary counting and geometric separation. Use strict coordinate inequalities to set rectangle boundaries between finite point coordinates, and use four disjoint triples among at least12 ordered points.

Definition 1. *Denote by $H_{\geq m}$ or simply H_m the hypergraph obtained from H by deleting all hyperedges of size smaller than m, and denote by $H_{=m}$ the hypergraph consisting of the hyperedges of H with size exactly m. Similarly, for a hypergraph family $\mathcal{H}$ let $\mathcal{H}_m = \mathcal{H}_{\geq m} = \{H_{\geq m} : H \in \mathcal{H}\}$ and $\mathcal{H}_{=m} = \{H_{=m} : H \in \mathcal{H}\}$.*

Definition 2. *Let $H = (V, E)$ be a hypergraph, a $U \subset V$ vertex set $U \subset V$ is a c-shallow hitting set for a positive integer c if for every hyperedge $e \in E$: $1 \leq |e \cap U| \leq c$.*

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Definition 3. *Denote by $\mathcal{B}$ the hypergraph family which consists of those hypergraphs $H = (V, E)$, for which $V \subset \mathbb{R}^2$ is finite, and the edges are induced by bottomless rectangles: every $e \in E$ edge can be written in the form $e = \{(x, y) \in V : x_0 < x < x_1, y < y_0\}$ for some $x_0, x_1, y_0 \in \mathbb{R}$.*

Notation. Directions in the plane are denoted by north, west, south, and east.

Theorem 1. *Let $m \geq 12$, then there is a member of $\mathcal{B}_{=m}$ which does not have a 3-shallow hitting set.*

Proof of Theorem 1. Take $\mathcal{B}_{=m}$ with $m \geq 12$. We will show that this family has an element with no 3-shallow hitting set. First take m points, $P = \{p_1, \ldots, p_m\}$ along a horizontal line, and above each of them, take the construction on Figure 1.

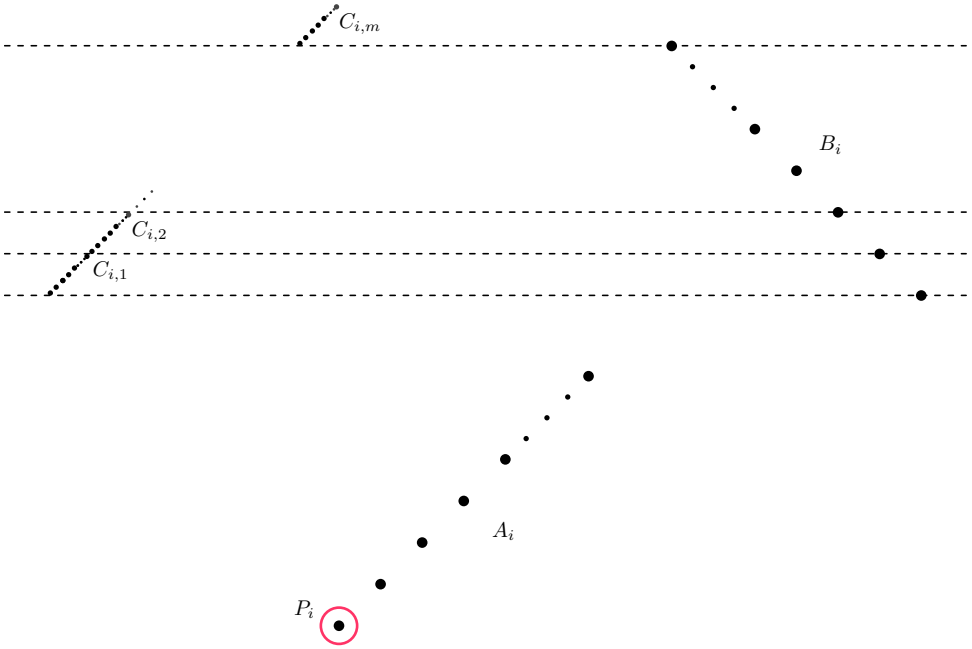


FIGURE 1. Construction for Theorem 1.

Starting from p_i we have A_i on a diagonal line consisting of $m-2$ points (including p_i) with increasing x and y coordinates, to northwest from A_i we have $B_i = \{b_i^1, \ldots, b_i^m\}$ on a diagonal line with increasing y, but decreasing x coordinates. To the west from A_i, we have m^2 points in a diagonal again with increasing x and y coordinates, consisting of m smaller groups $C_{i,j} = \{c_{i,j}^1, \ldots, c_{i,j}^m\}$ (for $j = 1, \ldots, m$) of size m, with $C_{i,j}$ being vertically between the b_i^j and b_i^{j+1}. More precisely, let $(x_1, y_1) \in A_i$, $(x_2, y_2) = b_i^k \in B_i$, and $(x_3, y_3) \in C_{i,j}$. Then $x_3 < x_1 < x_2$, $y_1 < y_2$, $y_1 < y_3$, and we have $y_2 < y_3$ exactly when $j \geq k$.

Let H be the hypergraph on this set which contains all m-sets contained in a bottomless rectangle as hyperedges. Now for the sake of contradiction suppose that this hypergraph has a 3-shallow hitting set.

There is a point in P which is in the hitting set, because these vertices form a hyperedge. Now we separate two cases:

Case 1. A_i contains at least one further point from the hitting set besides P. B_i consists of m points, and can be separated by a bottomless rectangle, so it has a point from the hitting set, say b_i^j. We can choose an axis-parallel rectangle that

contains exactly b_i^j from B_i. Also in $C_{i,j}$ there must be a point from the hitting set, so we can choose the rectangle in a way that its top left corner contains exactly this point from $C_{i,j}$, say $c_{i,j}^l$. Now we got a bottomless rectangle with m points in it, $m-2$ from A_i with at least two points from the hitting set, and b_i^j and $c_{i,j}^l$, both latter two from the hitting set. This is 4 in total; thus, the hitting set is not 3-shallow, a contradiction.

Case 2. A_i does not contain any more points from the hitting set (apart from p_i). Now we can take a bottomless rectangle which contains the $m-3$ rightmost points of A_i and any 3 neighbouring points from B_i. This implies that any 3 neighbouring points in B_i contain a point from the hitting set, which means B_i contains at least 4 points from the hitting set if $m \geq 12$, which is a contradiction. □